Problem Set 26

Rotation at Work: Volumes, Lengths, and Surfaces

November 10, 2026

Problem 1

(a)

$$\begin{aligned} F &= \int_0^R 2\pi r\, v(r)\, dr \\ &= 2\pi k \int_0^R rR^2 - r^3\, dr \\ &= 2\pi(\frac{r^2}{2}R^2 - \frac{r^4}{4})\Big|_0^R \\ &= \boxed{\frac{\pi k}{2}R^4} \end{aligned}$$

(b)

$$\begin{aligned} F/\pi R^2 &= \frac{1}{2}kR^2 \\ v(0) &= kR^2 \end{aligned}$$

(c)

$$\begin{aligned} 1 - 0.9^4 = 0.3439 &\implies 34.49\% \\ \frac{1}{0.9^4} \approx 1.524 &\implies 52.4\% \end{aligned}$$

Problem 2

(c)

$$\begin{aligned} S &= 2\pi b^2 + 2\pi ab\frac{\arcsin e}{e} \\ \lim_{e\to 0} S &= 2\pi b^2 + 2\pi ab \\ &= 2\pi b(a+b) \\ &= \boxed{4\pi a^2} \end{aligned}$$

The surface area approaches that of a sphere.

(d)

$$V = \frac{4}{3}\pi ab^2 = \frac{4}{3}\pi \cdot 2.85 \cdot 2.2^2 \approx 57.78 \text{ cm}^3$$
$$S = 2\pi \cdot 2.2^2 + 2\pi \cdot 2.85 \cdot 2.2 \cdot \frac{\arcsin e}{e} \approx 73.09 \text{ cm}^2$$
$$V \approx 73.09 \cdot 0.035 \approx 2.56 \text{ cm}^3$$

(e)

$$\frac{4}{3}\pi r^3 = 57.78 \implies r = \left(\frac{3 \cdot 57.78}{4\pi}\right)^{\frac{1}{3}}$$
$$S = 4\pi r^2 \approx 72.28 \text{ cm}^2$$
$$\frac{73.09 - 72.28}{72.28} \approx 1.12\%$$

Problem 3

(a)

$$dV = \frac{4}{3}\pi\left((r + dr)^3 - r^3\right)$$
$$= \frac{4}{3}\pi(3r^2\,dr + 3r\,d^2r + d^3r)$$
$$= \boxed{4\pi r^2}$$

(b)

$$\frac{1}{\pi a^3}e^{-2r/a}\,dv = \frac{4}{a^3}r^2e^{-2r/a}\,dr$$
$$P = \int_0^{ka} \frac{4}{a^3}r^2e^{-2r/a}\,dr$$
$$= \begin{bmatrix} u = r^2 & dv = e^{-2r/a}\,dr \\ du = 2r\,dr & v = -\frac{a}{2}e^{-2r/a} \end{bmatrix}$$
$$-\frac{4}{a^3} \cdot \frac{a}{2} \cdot r^2e^{-2r/a}\Big|_0^{ka} + \frac{4}{a^2}\int_0^{ka} re^{-2r/a}\,dr$$
$$\begin{bmatrix} u = r & dv = e^{-2r/a}\,dr \\ du = dr & v = -\frac{a}{2}e^{-2r/a} \end{bmatrix}$$
$$= -2k^2e^{-2k} - \frac{4}{a^2} \cdot \frac{a}{2}re^{-2r/a}\Big|_0^{ka} + \frac{4}{a^2} \cdot \frac{a}{2}\int_0^{ka} e^{-2r/a}\,dr$$
$$= -2k^2e^{-2k} - 2ke^{-2k} - \frac{2}{a} \cdot \frac{a}{2}e^{-2r/a}\Big|_0^{ka}$$
$$= \boxed{1 - e^{-2k}(2k^2 + 2k + 1)}$$

(c)

$$P(1) = 1 - 5e^{-2}$$
$$P(2) = 1 - 13e^{-4}$$
$$P(3) = 1 - 25e^{-6}$$
$$\lim_{k\to\infty} P(k) = 1$$

(d)

$$p = \frac{4}{a^3}r^2e^{-2r/a}$$
$$p' = \frac{4}{a^3}(2re^{-2r/a} - \frac{2}{a}r^2e^{-2r/a})$$
$$p' = 0 \implies r(1 - \frac{r}{a}) = 0 \implies r = \boxed{a}$$

The most likely distance is not $r = 0$, as the volume at that value is simply negligible.

Problem 4

(a)

$$x^2 + (y - r)^2 = r^2 \implies x^2 = 2ry - y^2$$
$$V = \pi \int_0^h 2ry - y^2\, dy$$
$$= \pi(ry^2 - \frac{y^3}{3})\Big|_0^h$$
$$= \boxed{\pi h^2(r - \frac{h}{3})}$$

(b)

$$V = \pi \cdot 10^2 \cdot \frac{1}{6} \cdot 10 = \frac{500\pi}{3}$$
$$\pi h^2(10 - \frac{h}{3}) = 100 \implies h = 2.797 \text{ cm}$$
$$\pi h^2(10 - \frac{h}{3}) = 250 \implies h = 4.850 \text{ cm}$$
$$\pi h^2(10 - \frac{h}{3}) = 400 \implies h = 6.843 \text{ cm}$$

(c)

$$V = \pi r h^2 - \frac{\pi}{3}h^3$$
$$V' = 2\pi rh - \pi h^2$$
$$V'' = 2\pi r - 2\pi h$$
$$V'' = 0 \implies \boxed{h = r}$$

Problem 5

(a)

$$x = 6 \implies y = 3$$
$$y = \frac{x^2}{12} \implies y' = \frac{1}{6}x$$
$$V = 2\pi \int_0^6 x\sqrt{1 + \frac{1}{36}x^2}\, dx$$
$$= \begin{bmatrix} u = \frac{1}{36}x^2 \\ du = \frac{1}{18}x\, dx \end{bmatrix} 36\pi \int_0^1 \sqrt{1+u}\, du$$
$$= 36\pi \cdot \frac{2}{3} \cdot (1+u)^{\frac{3}{2}}\Big|_0^1$$
$$= \boxed{24\pi(2\sqrt{2}-1)}$$

(b)

$$V = \pi \int_0^3 12y\, dy$$
$$= 6\pi y^2\Big|_0^3$$
$$= \boxed{54\pi}$$
$$V = 6^2 \cdot \pi \cdot 3 = 108\pi$$

(c)

$$x^2 + (y-r)^2 = r^2 \implies x^2 = 2ry - y^2$$
$$V = l\int_0^h 2\sqrt{2ry - y^2}\, dy$$
$$= l\int_0^h 2\sqrt{r^2 - (y-r)^2}\, dy$$
$$= l\int_0^{\arccos\frac{r-h}{r}} 2r\cos\theta \cdot r\sin\theta\, d\theta$$
$$= lr^2\int_0^{\arccos\frac{r-h}{r}} 2\sin^2\theta\, d\theta$$
$$= lr^2\int_0^{\arccos\frac{r-h}{r}} 1 - \cos 2\theta\, d\theta$$
$$= lr^2(\theta - \frac{\sin 2\theta}{2})\Big|_0^{\arccos\frac{r-h}{r}}$$
$$= lr^2\arccos\left(\frac{r-h}{r}\right) - lr^2 \cdot \frac{r-h}{r} \cdot \frac{\sqrt{2rh-h^2}}{r}$$
$$= \boxed{l\left(r^2\arccos\left(\frac{r-h}{r}\right) - (r-h)\sqrt{2rh-h^2}\right)}$$

(d)

$$V = \pi \cdot 1 \cdot 4 = 4000\pi\, L$$
$$V(0.5) = 19.55\%$$
$$V = \pi \implies h = 0.596$$

The dipstick is wider near the middle and thinner near the top and the bottom.

Problem 6

(a)

$$\begin{aligned} I &= \int_0^R r^2 \cdot \rho \cdot 2\pi r h \, dr \\ &= 2\pi\rho h \cdot \frac{r^4}{4}\Big|_0^R \\ &= \frac{1}{2}(\pi\rho h R^2)R^2 \\ &= \boxed{\frac{1}{2}MR^2} \end{aligned}$$

(b)

$$dm = \rho \cdot 2\pi x \cdot 2\sqrt{R^2 - x^2}\, dx$$
$$\begin{aligned} I &= 4\pi\rho \int_0^R x^3\sqrt{R^2 - x^2}\, dx \\ &= \begin{bmatrix} u = R^2 - x^2 \\ du = -2x\, dx \end{bmatrix} 2\pi\rho \int_0^{R^2} (R^2 - u)\sqrt{u}\, du \\ &= 2\pi\rho(R^2 \cdot \frac{2}{3}u^{\frac{3}{2}} - \frac{2}{5}u^{\frac{5}{2}})\Big|_0^{R^2} \\ &= \frac{8}{15}\pi\rho R^5 \\ &= \boxed{\frac{2}{5}MR^2} \end{aligned}$$

(c)

$$dm = 2\pi R\sigma\, dy$$
$$\begin{aligned} I &= \int_{-R}^R (R^2 - y^2)2\pi R\sigma\, dy \\ &= 4\pi R\sigma \int_0^R R^2 - y^2\, dy \\ &= 4\pi R\sigma(R^2 y - \frac{y^3}{3})\Big|_0^R \end{aligned}$$

$$= \frac{8}{3}\pi R^4 \sigma$$
$$= \boxed{\frac{2}{3}MR^2}$$

(d)

$$Solid\ ball > Cylinder > Hollow\ ball$$

The answer solely depends on the moment of inertia, not the mass.

Problem 7

(a)

$$x = 3 \implies y = 2\sqrt{3}$$
$$y = \frac{2}{3}x^{\frac{3}{2}} \implies y' = \sqrt{x}$$
$$L = \int_0^3 \sqrt{1+x}\,dx$$
$$= \frac{2}{3}(1+x)^{\frac{3}{2}}\Big|_0^3$$
$$= \boxed{\frac{14}{3}}$$

(b)

$$S = 2\pi \int_0^3 x\sqrt{1+x}\,dx$$
$$= 2\pi \int_1^4 (u-1)\sqrt{u}\,du$$
$$= 2\pi \int_1^4 u^{\frac{3}{2}} - \sqrt{u}\,du$$
$$= 2\pi(\frac{2}{5}u^{\frac{5}{2}} - \frac{2}{3}u^{\frac{3}{2}})\Big|_1^4$$
$$= \boxed{\frac{232\pi}{15}}$$

(c)

$$V = 2\pi \int_0^3 x(2\sqrt{3} - \frac{2}{3}x^{\frac{3}{2}})\,dx$$
$$= 2\pi \int_0^3 2\sqrt{3}x - \frac{2}{3}x^{\frac{5}{2}}\,dx$$
$$= 2\pi(\sqrt{3}x^2 - \frac{2}{3}\cdot\frac{2}{7}x^{\frac{7}{2}})\Big|_0^3$$

$$= 2\pi(9\sqrt{3} - \frac{36\sqrt{3}}{7})$$
$$= \boxed{\frac{54\sqrt{3}}{7}\pi}$$

(d)

$$\frac{3}{2}y = x^{\frac{3}{2}} \implies x^2 = (\frac{3}{2}y)^{\frac{4}{3}}$$
$$V = \pi \int_0^{2\sqrt{3}} (\frac{3}{2}y)^{\frac{4}{3}}\, dy$$
$$= \pi \cdot \frac{3}{2}^{\frac{4}{3}} \cdot \frac{3}{7}y^{\frac{7}{3}}\Big|_0^{2\sqrt{3}}$$
$$= \boxed{\frac{54\sqrt{3}}{7}\pi}$$

(e)

$$V = \pi \cdot \frac{3}{2}^{\frac{4}{3}} \cdot \frac{3}{7}h^{\frac{7}{3}}$$
$$V(\frac{1}{2}h) = \frac{1}{2^{7/3}}V$$

The area is much less than 1/2 as the bowl is more narrow near the bottom.

Problem 8

(c)

$$h = \frac{2d}{\sqrt{3}} \implies d = \frac{\sqrt{3}h}{2}$$
$$V_{max} = \pi(\frac{d^2}{4}h - \frac{1}{16}h^3) = \frac{\pi h^3}{8}$$
$$V = \frac{\pi}{16}(3h^2 \cdot h(1+\varepsilon) - h^3(1+\varepsilon)^3)$$
$$= \frac{\pi h^3}{8}(1 - \frac{3}{2}\varepsilon^2 - \frac{1}{2}\varepsilon^3)$$
$$= \boxed{V_{max}(1 - \frac{3}{2}\varepsilon^2 - \frac{1}{2}\varepsilon^3)}$$